Expanding Hopf-Lifted Ricci-Type Flows and Planck-Boundary Curvature Control in Compact S^3 Sectors

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ABSTRACT
We develop a second-stage analytic model for curvature control in a compact S^3 sector. The starting point is a conservative Ricci-type-flow framework: for $\partial\tau g = -2\,\mathrm{Ric}(g) + h[g]$, bounded tensorial spectral filtering is a well-defined forcing mechanism but does not, by itself, prevent singularities. Curvature boundedness requires a pointwise coercive contribution at the level of the curvature evolution,

$$2\langle C^h(\mathrm{Rm}), \mathrm{Rm}\rangle \le -\gamma|\mathrm{Rm}|^3 + A|\mathrm{Rm}|^2 + B, \quad \gamma > C_3.$$

The Berger diagnostics sharpen the result. For $g_{a,c} = a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2$ on $S^3 \simeq SU(2)$, unforced Ricci flow rounds the anisotropy toward $c = a$ and then collapses the round/base/volume mode. Hopf-fibre collapse $c \to 0$ at fixed $a > 0$ has bounded curvature and is therefore not the singular mode. Constant-Λ trace forcing gives only the quadratic contribution $-4\Lambda|\mathrm{Rm}|^2$; fibre-only forcing changes sign at $u = (c/a)^2 = 12/11$ and cannot control the round mode; traceless/spectral shape forcing vanishes at the round configuration. The positive homogeneous result comes from the curvature-driven trace flow $\partial\tau g = -2\,\mathrm{Ric} + (2/3)Rg$. It preserves $V = a^2c$, drives u toward 1, and gives an explicit curvature bound when combined with a Planck lower boundary $R_{eff} = V^{\{1/3\}} \ge \ell P$: $|\mathrm{Rm}|^2 \le C(u_0)\ell P^{-4}$.

The new warped-S^3 diagnostic tests the only engineered cubic term that succeeded in the Berger reduction, $h = 2\lambda_1|\mathrm{Rm}|g$. For a rotationally symmetric metric $g = ds^2 + \psi(s)^2 gS^2$, the conformal variation $\delta R = -2\varphi R - 4\Delta\varphi$ shows that the full scalar-curvature evolution contains not only the retained damping term $-2\lambda_1|\mathrm{Rm}|R$ but also the derivative term $-4\lambda_1\Delta|\mathrm{Rm}|$, which the homogeneous reduction sets to zero. On sharpening neck profiles this discarded term remains of the same order as the retained term. Thus the engineered cubic coercivity is a symmetry-reduction artifact, not a full-PDE mechanism. The clean curvature-control mechanism presently established is therefore a boundary/constraint mechanism: trace/shape control together with a Planck no-flux lower boundary.

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I. INTRODUCTION

Ricci flow evolves a Riemannian metric by

$$\partial\tau\, g_{ij} = -2\, \mathrm{Ric}_{ij}(g) \quad (1)$$

In three dimensions it is a central analytic tool because singularities appear in a controlled geometric way, and Perelman's use of surgery gives an external continuation procedure through neck-like regions. That classical theorem is **not** being reproved here. The present question is different.

Suppose the spatial sector is already fixed and compact, with $M \simeq S^3$. A smooth metric flow on this fixed manifold cannot change topology. The hard question is not topological classification but analytic *curvature control*. If a physical model contains no external agent and no ambient space in which to cut, cap, and restart the evolution, then the analogue of surgery, if any, must be internalized. It must appear as a boundary condition, a constraint, a compensating sector, an additional geometric degree of freedom, or a change of reconstruction clock.

The first-stage spectral-filter framework isolated the analytic bottleneck. For a Ricci-type flow

$$\partial\tau\, g = -2\, \mathrm{Ric}(g) + h[g] \quad (2)$$

the curvature evolution contains a cubic reaction term:

$$(\partial\tau - \Delta)|\mathrm{Rm}|^2 \le C_3|\mathrm{Rm}|^3 + 2\langle C^h(\mathrm{Rm}), \mathrm{Rm}\rangle \quad (3)$$

Thus boundedness of h, even if h is a geometrically legitimate tensorial spectral filter, does **not** imply curvature control. A concrete forcing must satisfy or imply a pointwise coercive estimate strong enough to dominate the cubic term.

This paper develops the next test model. The Hopf fibration

$$S^1 \longrightarrow S^3 \simeq \mathrm{SU}(2) \longrightarrow S^2 \quad (4)$$

is a natural candidate because it separates a base-like two-dimensional sector from a fibre/phase/connection sector. The minimal test model is the Berger family

$$g_a{}^c = a^2(\sigma_1{}^2 + \sigma_2{}^2) + c^2\sigma_3{}^2 \quad (5)$$

where σ_3 is adapted to the Hopf fibre. This model gives a sharp answer. It shows that fibre collapse and curvature blow-up are *not* the same phenomenon. At fixed $a > 0$, the limit $c \to 0$ collapses the S^1 fibres but leaves the curvature bounded. The genuine blow-up occurs in the round/base/volume mode $a \to 0$. Therefore a fibre-only regularization mechanism attacks the wrong singular mode.

The missing homogeneous sector is the trace or expansion sector. A physical cosmological slice is not described by g_{ij} alone. In ADM variables the data include the second fundamental form K_{ij}, and for an expanding FLRW slice, $K_{ij} = -Hg_{ij}$. Thus expansion appears, in a Ricci-type reconstruction equation, as a conformal forcing $2\lambda g$. This is the main correction to the fibre-only programme.

The present update incorporates the Berger phase portrait, the coercive sign diagnostic, the numerical protocol, and a warped-S^3 hidden-derivative diagnostic. These additions do not change the Berger theorem: they clarify why the fibre-only programme is a near-no-go, why the trace/ADM expansion sector is the correct homogeneous variable, and why the engineered cubic term $h = 2\lambda_1|\mathrm{Rm}|g$ does not survive the first nonhomogeneous test.

II. CLAIMS AND NON-CLAIMS

The claims are intentionally narrow.

A. Established in this paper

(1) The Berger sphere formulas are derived with an explicit SU(2) normalization.
(2) Unforced Ricci flow on the Berger family rounds the anisotropy but collapses the round scale in finite time.
(3) $|\mathrm{Rm}|^2 = 4(16a^4 - 24a^2c^2 + 11c^4) / a^8$.
(4) The limit $c \to 0$ at fixed $a > 0$ has bounded curvature, while $a \to 0$ gives blow-up.
(5) Constant-Λ trace forcing contributes only $-4\Lambda|\mathrm{Rm}|^2$ in the Berger reduction; it does not produce a negative cubic term.
(6) A fibre-only term $h = \mu\, \sigma_3 \otimes \sigma_3$ has the wrong global sign structure and does not control the round mode.
(7) The curvature-driven trace term $h = (2/3)\,\mathrm{Rg}$ preserves Berger volume, drives $u = (c/a)^2 \to 1$, and yields a curvature bound under a Planck lower boundary.
(8) The full PDE problem remains open because homogeneous reductions remove derivative terms in $C^h(\mathrm{Rm})$.

In addition, the warped-S^3 tests expose the hidden derivative term $-4\lambda_1\Delta|\mathrm{Rm}|$ in the engineered forcing $h = 2\lambda_1|\mathrm{Rm}|g$. Frozen neck profiles show that this term is the same order as the retained cubic term, and the Ricci–DeTurck diagnostic shows dynamically that the engineered forcing delays and relocates, but does not remove, curvature concentration.
The unforced phase portrait and coercive diagnostic are included as reproducibility figures: they make visible that fibre collapse is the bounded-curvature direction and that the cubic obstruction is the round trace mode.

B. Not established

We do not claim:

(9) that Hopf fibration removes Ricci-flow singularities;
(10) that dark energy is fibre curvature;
(11) that dark matter is solved;
(12) that the Big Bang is Perelman's neck;
(13) that a bounded spectral filter is a high-frequency parabolic regularizer;
(14) that a Planck boundary is derived from microscopic quantum gravity;
(15) that normalized Ricci flow proves physical cosmological singularity resolution;
(16) that a result in the Berger minisuperspace proves the full nonhomogeneous PDE estimate.

We also do not claim that the warped-S^3 diagnostics prove a general nonexistence theorem for all possible forcings. They rule out the tested engineered cubic term as evidence for the full pointwise coercive inequality.

III. LOGICAL ROLE OF THE EXISTING MODULES

The research programme has four distinct modules. Confusing them weakens the mathematics.

A. Tensorial spectral filtering

The first module constructs tensorial spectral filters on a fixed compact S^3 sector. Given a reference metric $\dot{g}$ and a positive self-adjoint elliptic operator $L_{\dot{g}}$ on Sym^2T^*M, functional calculus gives tensorial band projectors $P^{\dot{g}}_n$.

For bounded weights w_n,

$$B^W_w\, h = -\Sigma\, w_n\, P^T_n\, h \quad (10)$$

is coordinate-invariant once $\dot{g}$ is fixed. This is a legitimate bounded forcing operator. It is **not** by itself a curvature-control theorem.

The key theorem of that module is conditional. If h[g] contributes to the curvature evolution through $C^h(\mathrm{Rm})$ and if

$$2\langle C^h(\mathrm{Rm}), \mathrm{Rm}\rangle \le -\gamma|\mathrm{Rm}|^3 + A|\mathrm{Rm}|^2 + B, \quad \gamma > C_3 \qquad (11)$$

at spatial maxima of |Rm|, then |Rm| stays bounded by scalar comparison. The open problem is to find or rule out concrete h's satisfying such an inequality.

B. Planck-boundary compact S^3 cosmology

The second module gives an effective boundary structure: $a \in [\ell_P, \infty)$, with $a = 0$ excluded. For closed FLRW data with $k = +1$:

$$K_{sl}^{RW} = 12[(\dot{H} + H^2)^2 + (H^2 + 1/a^2)^2] \qquad (13)$$

Therefore finite $H(t_P)$, finite $\dot{H}(t_P)$, and $a_{eff}(t_P) \ge \ell_P$ imply $K_{FLRW}(t_P) \le C\ell_P^{-4}$. This is a kinematic effective endpoint theorem.

C. Hopf-bundle holonomy

The third module concerns the standard spinor geometry $S^1 \to S^3 \subset \mathbb{C}^2 \to \mathbb{C}P^1 \simeq S^2$. Its role here is conceptual and geometric: a quotient description on the base can hide fibre/phase/connection data present on the total space. The Berry connection and its holonomy show that such data can be gauge-invariant.

D. Research-programme comments

The fourth module asks whether Perelman surgery has an internal analogue in a compact physical S^3 sector. The present paper turns that programme into a concrete test:

$$\text{external surgery} \rightsquigarrow \text{trace compensation} + \text{Hopf shape control} + \text{Planck boundary}$$

IV. RICCI-TYPE CURVATURE COMPARISON

THEOREM IV.1 (Conditional curvature comparison). Let $g(\tau)$ be a smooth solution of $\partial_\tau g = -2\,\mathrm{Ric}(g) + h[g]$ on a compact three-manifold M, $\tau \in [0, T]$. Suppose that at every spacetime point where |Rm| attains its spatial maximum at time τ, $2\langle C^h(\mathrm{Rm}), \mathrm{Rm}\rangle \le -\gamma|\mathrm{Rm}|^3 + A|\mathrm{Rm}|^2 + B$, $\gamma > C_3$. Then $K(\tau) = \sup|\mathrm{Rm}(g(\tau))|$ remains bounded on $[0, T]$.

Proof. Set $u(\tau) = K(\tau)^2$. At a spatial maximum of $|\mathrm{Rm}|^2$, the Laplacian term in (3) is non-positive in the upper-Dini derivative sense. Hence $D^+u \le -(\gamma - C_3)u^{3/2} + Au + B$. The right-hand side is negative for sufficiently large u. Scalar comparison gives an absorbing upper level. □

COROLLARY IV.2 (Traceless damping misses the round mode). Any forcing depending only on $E = \mathrm{Ric} - (R/3)g$ vanishes on round metrics. It cannot, by itself, prevent round shrinking curvature blow-up.

This corollary is one of the hinges of the present article. It means that anisotropy damping and spectral damping of E may improve shape, but the round trace mode requires a separate mechanism.

V. RICCI FLOW VERSUS COSMOLOGICAL EXPANSION

Ricci flow is intrinsic. It evolves only the Riemannian metric of a spatial manifold. Physical cosmological evolution, in the ADM decomposition, uses both the spatial metric and the extrinsic curvature: (g_{ij}, K_{ij}).

$$\partial_t\, g_{ij} = -2NK_{ij} + \mathcal{L}_\beta\, g_{ij} \quad (21)$$

For a homogeneous expanding FLRW slice with $N = 1$ and $\beta = 0$: $K_{ij} = -Hg_{ij}$, so $\partial_t g_{ij} = 2Hg_{ij}$. Thus the minimal Ricci-type reconstruction equation that includes expansion is:

$$\partial\tau\, g = -2\,\mathrm{Ric} + 2\lambda g + h_\mathrm{shape} + h_\mathrm{boundary} + \cdots \quad (23)$$

Here λ is not automatically the physical Hubble rate. It is a reconstruction-clock coefficient representing the trace sector. A constant $\lambda = \Lambda$ is too weak for high-curvature coercivity; a curvature-driven λ may have the correct homogeneity but creates new analytic terms in the full PDE.

Remark V.1. The distinction between Ricci-flow time τ and physical time t is essential. Equation (23) is a Ricci-type reconstruction model. It is not the Einstein equation written in disguise.

VI. HOPF/BERGER GEOMETRY ON S^3 ≅ SU(2)

A. Convention

Let $\{e_i\}$ be a left-invariant frame on SU(2) satisfying $[e_i, e_j] = 2\varepsilon_{ijk}e_k$. Let $\{\sigma_i\}$ be the dual coframe with $d\sigma^1 = -2\sigma^2 \wedge \sigma^3$ (and cyclic permutations). The round metric is $g_{rd} = \sigma^2{}_1 + \sigma^2{}_2 + \sigma^2{}_3$, with $\mathrm{Ric} = 2g_{rd}$.

The Berger family is

$$g_a{}^c = a^2(\sigma^2{}_1 + \sigma^2{}_2) + c^2\sigma^2{}_3, \quad a > 0, c > 0. \quad (27)$$

Here a is base-like, c is fibre-like, and $x = a^2$, $z = c^2$, $u = z/x = (c/a)^2$. The spatial volume is proportional to $V = a^2c = x^{3/2}u^{1/2}$, and the effective volume radius is $R_{\mathrm{eff}} = V^{1/3}$.

B. Ricci Tensor

With orthonormal frame $\theta^1 = a\sigma^1$, $\theta^2 = a\sigma^2$, $\theta^3 = c\sigma^3$, the Ricci eigenvalues are:

$$r_1 = r_2 = 2(2 - u) / x, \quad r_3 = 2u / x. \quad (31)$$

Therefore the scalar curvature is $R = (8 - 2u) / x$. At $a = c = 1$ (round point), $u = 1$ and $r_1 = r_2 = r_3 = 2$, as required.

C. Curvature Invariants

In dimension three the Riemann tensor is determined by Ricci: $|\mathrm{Rm}|^2 = 4|\mathrm{Ric}|^2 - R^2$. This gives

$$K^B(x, u) := |\mathrm{Rm}|^2 = 4(16 - 24u + 11u^2) / x^2 \qquad = 4(16a^4 - 24a^2c^2 + 11c^4) / a^8. \quad (35)$$

At the round point $u = 1$: $|\mathrm{Rm}|^2 = 12/x^2$.

VII. UNFORCED BERGER RICCI FLOW

For $\partial\tau g = -2\,\mathrm{Ric}(g)$, the coefficients $x = a^2$ and $z = c^2$ satisfy

$$\dot{x} = -8 + 4u, \quad \dot{z} = -4u^2. \quad (38)$$

Consequently $\dot{u} = 8u(1-u)/x$. Thus $u = 1$ is the round fixed ratio and is an attractor in shape. However, the volume is not preserved: $d/d\tau\ (\log V) = -R = (-8 + 2u)/x$. Near the round trajectory $u = 1$, $d/d\tau\ (\log V) = -6/x < 0$. The unforced flow rounds and then collapses in finite time.

A. The Two Collapse Regimes

a. Fibre collapse

If $c \to 0$ with $a > 0$ fixed, then $u \to 0$ and $|\mathrm{Rm}|^2 \to 64/a^4$. This is bounded for fixed a. Geometrically S^3 collapses along the Hopf fibres to a round S^2 with bounded sectional curvature. The reduced description degenerates, but the curvature of the total Berger geometry need not blow up.

b. Round/base/volume collapse

If $a \to 0$ while u remains bounded away from infinity, then $|\mathrm{Rm}|^2 \sim C(u)/a^4 \to \infty$. This is the true homogeneous singular mode.

> **PROPOSITION VII.1 (Fibre collapse is not the singular Berger mode).** In the Berger family (27), the limit $c \to 0$ with $a > 0$ fixed has bounded $|\mathrm{Rm}|^2$. The unforced Ricci-flow blow-up occurs through collapse of the scale a, equivalently through volume/trace collapse after the ratio u has been driven toward 1.

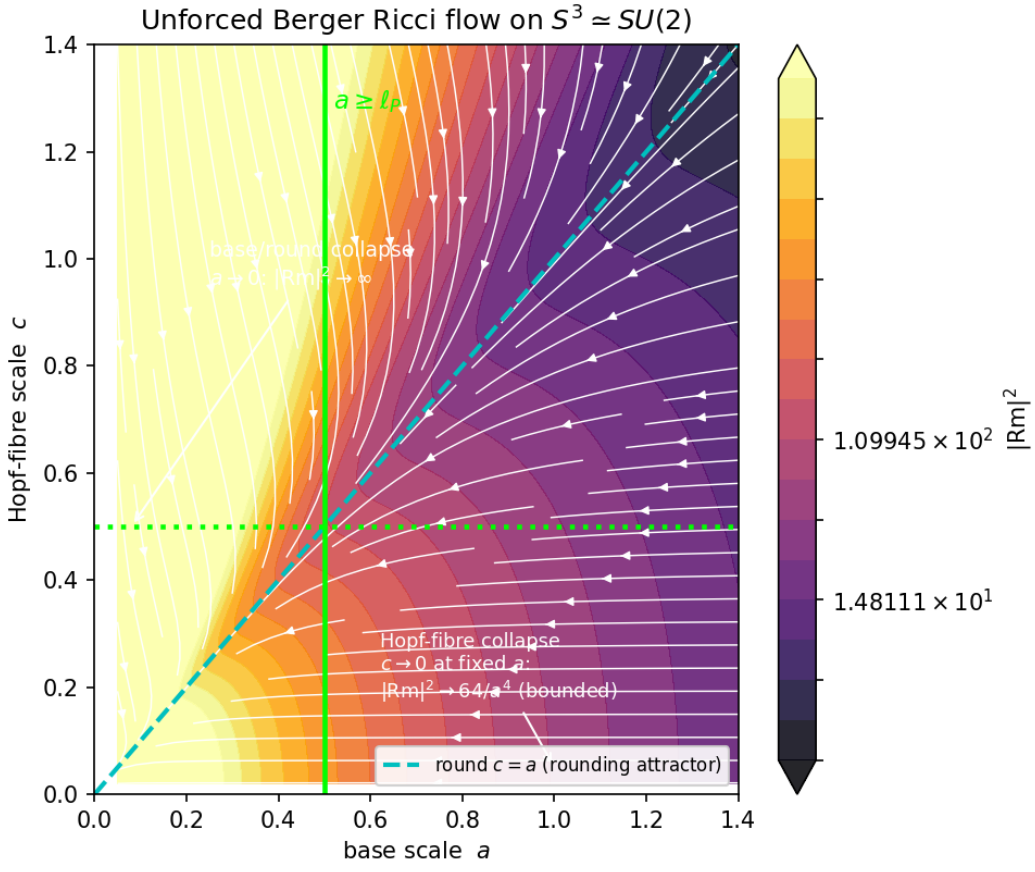


Figure 1. Unforced Berger Ricci flow on S^3 ≅ SU(2). The vector field rounds trajectories toward $c = a$, but the subsequent singular direction is the base/round collapse $a \to 0$. Hopf-fibre collapse $c \to 0$ at fixed a remains bounded-curvature. The green line marks a representative Planck lower bound $a \geq \ell P$; it is a boundary/constraint, not a smooth forcing.

VIII. CANDIDATE FORCING MECHANISMS IN THE BERGER REDUCTION

The finite-dimensional test is not a proof of the PDE theorem, but it is a useful algebraic laboratory. Let $K_B = |\mathrm{Rm}(g_{a,c})|^2$. For each forcing we compute its contribution to $dK_B/d\tau$.

A. Constant dark-energy-like trace forcing

Let $h_\Lambda = 2\Lambda g$. Since K_B is homogeneous of degree −2 in (x, z), Euler's identity gives:

$$dK^B/d\tau\,|_\Lambda \;=\; -4\Lambda\, K^B. \quad (46)$$

This is damping of order K_B, i.e., order $|\mathrm{Rm}|^2$. It does **not** dominate the cubic reaction scale $|\mathrm{Rm}|^3$.

B. Curvature-dependent trace forcing

Let $h_\lambda = 2\lambda(g)g$. If $\lambda = \lambda_1|\mathrm{Rm}|$, then $dK_B/d\tau = -4\lambda_1|\mathrm{Rm}|^3$, which has the right homogeneity. A natural homogeneous choice is

$$h^B = (2/3)\, R\, g. \quad (50)$$

This is precisely the volume-normalizing trace term for a homogeneous three-dimensional flow.

C. Fibre-restoring forcing

Let $h_{fib} = \mu\, \sigma_3 \otimes \sigma_3$, so $\dot{x}|_{fib} = 0$, $\dot{z}|_{fib} = \mu$. From (35):

$$dK^B/d\tau\,|_{fib} \;=\; \mu \cdot 8(11u - 12)\,/\,x^3. \quad (53)$$

For positive μ, this term is negative only when $u < 12/11$. It becomes positive for sufficiently stretched fibres. More importantly, it has no mechanism to prevent $a \to 0$ along the round direction. **Fibre protection is not volume protection.**

D. Traceless and spectral shape forcing

A traceless-Ricci forcing built from $E = \mathrm{Ric} - (1/3)Rg$ vanishes when $u = 1$. Therefore it cannot oppose the round shrinking mode. The bounded spectral tensor filter of the first-stage paper is a legitimate tensorial operation if built from a reference-metric elliptic operator on Sym^2T^*M. But if it acts on E, it shares the same round-mode defect: $E(g_{round}) = 0$.

E. Summary of the no-go part

(17) Constant $2\Lambda g$ contributes $-4\Lambda K^B$. Too weak to supply cubic coercivity.

(18) Curvature-dependent trace forcing $2\lambda|\mathrm{Rm}|g$ has the correct homogeneity, but is engineered unless derived from a geometric or physical principle, and hides derivative terms in the full PDE.

In the warped-S^3 diagnostics this same engineered term acquires the hidden contribution $-4\lambda_1\Delta|\mathrm{Rm}|$, which is not lower order near a neck; in Ricci–DeTurck dynamics it delays and relocates, but does not remove, blow-up.

(19) Fibre-only forcing $\mu\sigma^2_3$ protects c only. Wrong sign for $u > 12/11$ and does not protect a.

(20) Traceless E-forcing vanishes at u = 1. Misses round collapse.

(21) Bounded spectral E-filtering is legitimate tensor forcing, but controls shape at best, not volume.

(22) A Planck boundary is a clean lower-domain mechanism. It is a condition or constraint, not a smooth forcing.

Numerical and algebraic diagnostics make the sign structure explicit.

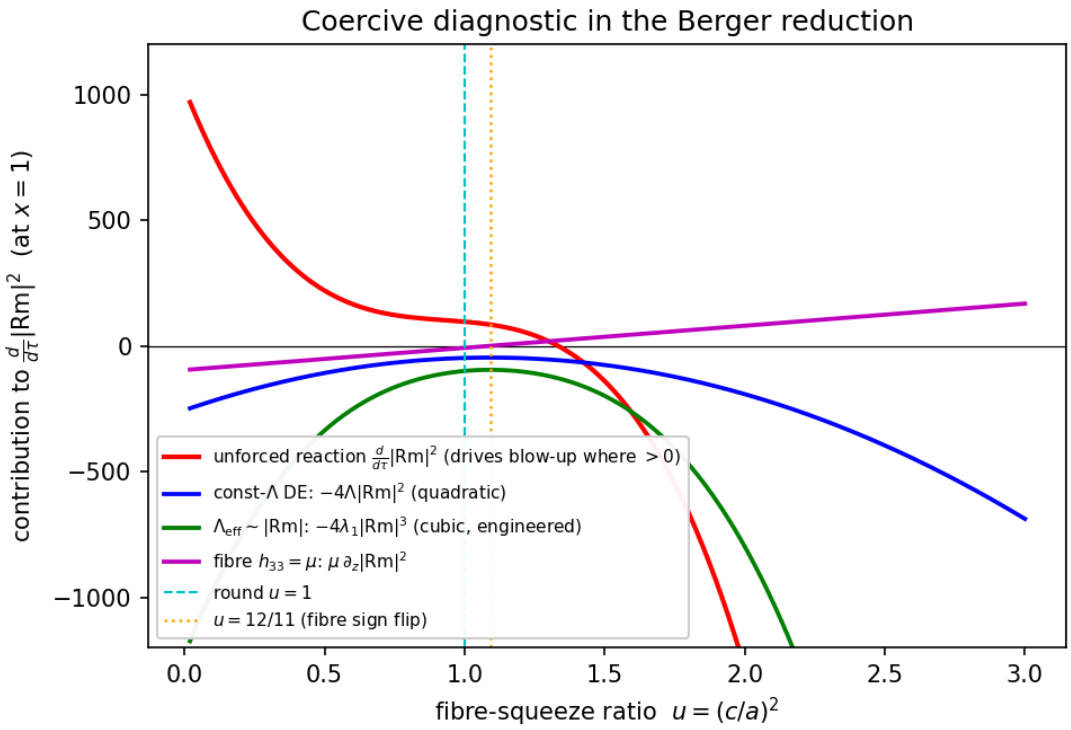


Figure 2. Coercive diagnostic in the Berger reduction. The unforced reaction is the positive cubic obstruction near u = 1. Constant Λ gives only quadratic damping; Λeff ∼ |Rm| gives a cubic but is engineered and hides full-PDE derivative terms; fibre forcing changes sign at u = 12/11.

Proposition VIII.1 (Berger no-go diagnostic). In the left-invariant Berger reduction, no bounded forcing of pure constant-trace type, pure fibre type, or traceless/spectral shape type supplies a negative cubic term controlling the round-collapse direction. Constant trace forcing is quadratic; fibre forcing is sign-limited and does not protect a; traceless/spectral shape forcing vanishes at u = 1. The proof is the Euler-homogeneity computation of Appendix B and is illustrated in Figure 2.

IX. POSITIVE RESULT: THE EXPANDING NORMALIZED BERGER FLOW

We now add the trace/expansion term that the previous no-go mechanisms lack:

$$\partial\tau\, g = -2\,\mathrm{Ric} + (2/3)\,R\,g. \quad (56)$$

In the homogeneous Berger sector, R is spatially constant, so (56) is the volume-normalized Ricci flow.

A. ODE System

Combining (38) with the forcing contribution and using $R = (8 - 2u)/x$, we get

$$\dot{x} = (8/3)(u - 1), \quad \dot{z} = (16/3)\,u(1 - u). \quad (57)$$

The ratio evolves by the same equation as before: $\dot{u} = 8u(1 - u)/x$.

PROPOSITION IX.1 (Volume preservation). For the normalized Berger flow (56), $V = a^2c = x\sqrt{z}$ is constant.

Proof. Using (57): $d/d\tau\,(\log V) = \dot{x}/x + (1/2)\dot{z}/z = 8(u-1)/(3x) + 8u(1-u)/(3ux) = 0$.

PROPOSITION IX.2 (Round ratio attraction). For every solution of (56) with $u(0) > 0$, the ratio $u(\tau)$ remains between $u(0)$ and 1, and $u(\tau) \to 1$ as $\tau \to \infty$.

Proof. Equation (58) gives $\dot{u} > 0$ for $0 < u < 1$ and $\dot{u} < 0$ for $u > 1$. Thus u is monotone toward 1 and cannot cross 1. Since V is fixed, $x = V^{2/3}u^{-1/3}$ stays finite and positive as long as u stays in a compact subinterval of $(0, \infty)$. The only possible limit compatible with the sign of $\dot{u}$ is $u = 1$. □

B. Curvature Evolution in the Normalized Berger Flow

Differentiating (35) using (57), one obtains

$$dK^B/d\tau = -(256/3x^3)(u-1)^2(11u-4). \quad (60)$$

This expression is not everywhere non-positive: for $0 < u < 4/11$ it is positive. Monotonicity of K_B is therefore not the correct theorem. **Boundedness is the correct theorem.** Since V is constant, $x = V^{2/3}u^{-1/3}$, and substituting into (35):

$$K^B = 4(16 - 24u + 11u^2)u^{2w3} / R^4_eff. \quad (62)$$

THEOREM IX.3 (Berger trace-compensation bound). Let $g_{a,c}(\tau)$ solve $\partial\tau g = -2\,\mathrm{Ric} + (2/3)Rg$ in the Berger family with $a_0 > 0$, $c_0 > 0$. Set $u_0 = (c_0/a_0)^2$ and $R_eff = (a_0^2 c_0)^{\wedge}\{1/3\}$. Then R_eff is constant and $u(\tau)$ stays in $I(u_0) = [\min\{u_0,1\}, \max\{u_0,1\}]$. Hence

$$|Rm(g(\tau))|^2 \le C_B(u_0)\, R_eff^{\wedge}\{-4\},$$

$$C_B(u_0) = \max_\{u \in I(u_0)\}\, 4(16 - 24u + 11u^2)u^{\wedge}\{2/3\}.$$

If $R_eff \ge \ell P$, then $|Rm(g(\tau))|^2 \le C_B(u_0)\ell P^{\wedge}\{-4\}$.

Remark IX.4. The theorem is a genuine positive result in the Hopf/Berger reduction. It says that the round collapse can be removed by adding the correct trace/expansion sector and imposing a lower effective volume scale.

Remark IX.5. The theorem does not prove a full pointwise coercive PDE estimate. It uses spatial homogeneity, so all derivative terms in $C^h(Rm)$ vanish. Nonhomogeneous trace forcing is a harder problem.

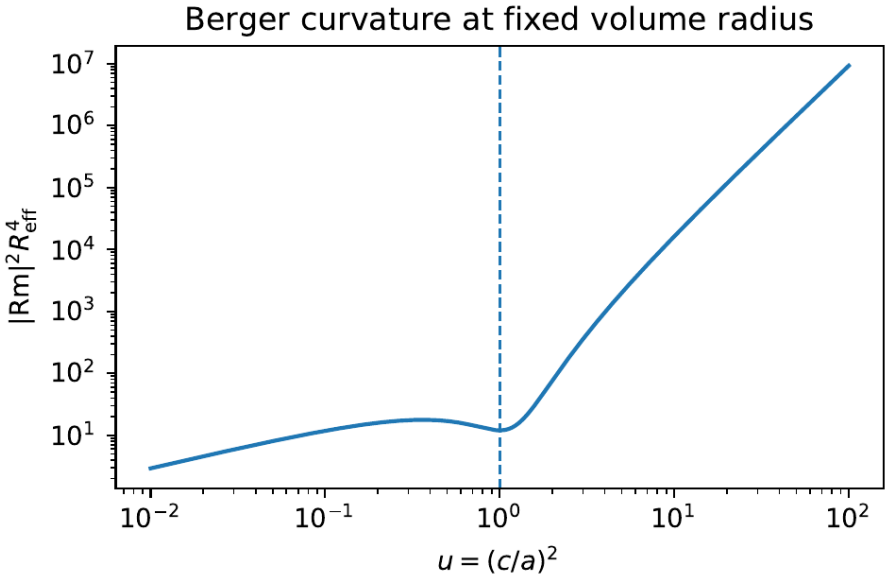


Figure 3. Fixed-volume Berger curvature. Under the normalized trace flow $V = a^2c$ is preserved and u moves monotonically toward 1. The theorem proves boundedness rather than monotonic decay: curvature can increase on part of the fibre-squeezed side, but remains bounded on the compact u-interval determined by the initial data.

X. PLANCK BOUNDARY AS INTERNALIZED SURGERY

Perelman surgery is an external analytic-topological continuation method. It removes high-curvature neck regions, inserts caps, and restarts the flow. In a compact physical S^3 sector there is no external operator and no outside ambient space. Therefore the replacement cannot literally be cut-and-paste surgery. The effective replacement is a boundary or constraint.

The Planck-boundary module supplies precisely such a boundary:

$$R_eff \geq \ell_p, \quad \text{or stronger variants such as } a \geq \ell_p,\ c \geq \ell_p. \qquad (66)$$

For the Berger trace-compensated flow, the natural variable is $R_{eff} = (a^2c)^{1W3}$. The theorem above shows that the boundary $R_{eff} \geq \ell_P$ is sufficient in the volume-normalized homogeneous sector.

A hard boundary condition is cleaner than a divergent forcing term. In the minisuperspace quantum model one chooses a Hilbert space $L^2([\ell_P, \infty), a\,da)$ and a self-adjoint endpoint condition, so no probability flux is lost into $0 < a < \ell_P$. The Berger analogue should be a self-adjoint boundary condition on an anisotropic minisuperspace with variables such as (V, u) or (log V, log u).

XI. REDUCED VERSUS LIFTED DEGENERATION

The statement that "ordinary Ricci flow may not see all degrees of freedom" must be corrected. Ricci flow sees the full metric on the manifold on which it is defined. The loss occurs only if the model first quotients, projects, or coarse-grains the system.

In the Hopf case $S^1 \to S^3 \to S^2$, a reduced S^2 description can lose:

- fibre radius;
- connection one-form;
- holonomy data;
- O'Neill tensor contributions;
- boundary or asymptotic charge data associated with the lifted geometry.

The Berger family makes this distinction explicit. The limit $c \to 0$ is a collapse of S^3 to S^2 with bounded curvature upstairs. The **safe version** of the lifted-flow idea is: a reduced flow may be incomplete relative to a lifted total-space geometry.

XII. CONNECTION-CURVATURE AND DARK-SECTOR INTERPRETATION

The Hopf connection θ has curvature $F = d\theta$. In a Riemannian submersion or Kaluza–Klein reduction, total-space curvature decomposes schematically into base curvature, fibre scale, and connection-curvature terms. This is the precise geometric place where a reduced observer might describe fibre/connection effects as an effective stress or pressure sector.

For the present article, this remains a modelling direction, not a theorem. A defensible dark-sector ansatz is

$$h_comp = h_trace + h_shape + h_fibre + h_boundary + h_nonlocal. \qquad (67)$$

The trace term is the natural dark-energy-like component because it affects volume expansion.

The disciplined claim is therefore: *the dark sector may be tested as an effective internal compensation sector.* The unsupported claim would be: *dark energy is Hopf fibre curvature.* We do not make the second claim.

XIII. Warped-S^3 Test of the Engineered Cubic Forcing

The Berger reduction is homogeneous: $|\mathrm{Rm}|$ is constant on each spatial slice and $\Delta|\mathrm{Rm}| = 0$. It is therefore the wrong environment for deciding whether the engineered cubic forcing A', namely $h = 2\lambda_1|\mathrm{Rm}|g$, survives the full curvature equation. The first honest nonhomogeneous test is the rotationally symmetric warped sector on S^3, the same symmetry class in which Ricci flow develops neckpinch behaviour.

A. Warped curvature dictionary

Let
$g = ds^2 + \psi(s)^2\, g_{S^2}$. (68)
In arclength gauge the two sectional curvatures are
$K_0 = -\psi_{ss}/\psi, \quad K_1 = (1 - \psi_s^2)/\psi^2$. (69)
Here K_0 is the mixed radial-spherical curvature and K_1 is the curvature of the spherical orbits. Hence
$R = 4K_0 + 2K_1, \quad |\mathrm{Rm}|^2 = 4(2K_0^2 + K_1^2)$. (70)
For $\psi(s) = \sin s$, one obtains $K_0 = K_1 = 1$, so $R = 6$ and $|\mathrm{Rm}|^2 = 12$. This is the round S^3 check.

B. The derivative term hidden by homogeneity

For a conformal variation $\delta g = 2\varphi g$ in dimension three,
$\delta R = -2\varphi R - 4\Delta\varphi$. (71)
Putting $\varphi = \lambda_1|\mathrm{Rm}|$ gives the scalar-curvature contribution of $h = 2\lambda_1|\mathrm{Rm}|g$:
$\partial_\tau R = \Delta R + 2|\mathrm{Ric}|^2 - 2\lambda_1|\mathrm{Rm}|R - 4\lambda_1\Delta|\mathrm{Rm}|$. (72)
The Berger reduction retains the algebraic term $-2\lambda_1|\mathrm{Rm}|R$ and kills the derivative term $-4\lambda_1\Delta|\mathrm{Rm}|$. That is precisely the term denoted schematically by $\nabla^2 h$ in the full curvature comparison. It is not an optional correction; it is the term that decides whether the engineered cubic is a real PDE mechanism.
At a curvature maximum on a neck one typically has $\Delta|\mathrm{Rm}| < 0$. Thus $-4\lambda_1\Delta|\mathrm{Rm}|$ is positive there and can feed the scalar curvature exactly where the homogeneous model suggested damping. Even when the scalar curvature on an acute neck changes sign, the essential point is order, not sign: the derivative term is not lower order.

C. Ricci–DeTurck dynamic diagnostic

The frozen-profile calculation already shows that the derivative term removed by homogeneity is of the same order as the algebraic term kept by the Berger reduction. A dynamical test asks the sharper question: does the engineered forcing actually prevent a neckpinch when the warped-S^3 flow is evolved?
We use a rotationally symmetric Ricci–DeTurck formulation with round background $g_{\mathrm{bar}} = dx^2 + \sin^2 x\, gS^2$. The system is strictly parabolic in the variables (φ,ψ), preserves the round solution to numerical precision, and resolves unforced neckpinch behaviour. The same solver is then run with $h = 2\lambda_1|\mathrm{Rm}|g$.
The outcome is negative in the relevant sense. The engineered term can hold the central neck open and delay the first concentration, but the curvature peak is displaced rather than removed. In the diagnostic run the singularity is therefore not regularized; it is relocated. This is precisely what the hidden term $-4\lambda_1\Delta|\mathrm{Rm}|$ suggested analytically.
The frozen-profile ratios are retained in Appendix D as a scale check. The main figure below records the dynamic Ricci–DeTurck result.

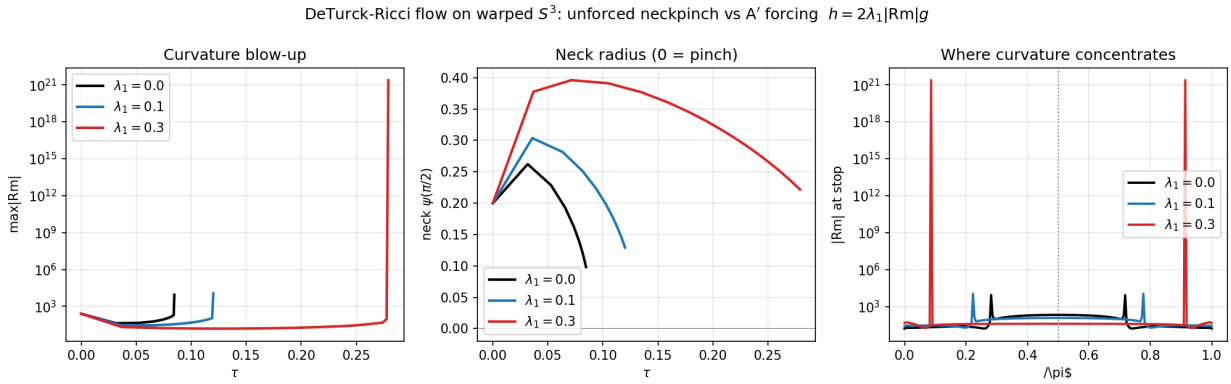


Figure 4. Ricci–DeTurck warped-S^3 dynamic diagnostic. The engineered forcing $h = 2\lambda_1|\mathrm{Rm}|g$ holds the central neck open and delays the first concentration, but the curvature peak is displaced rather than removed. Left: maximal curvature as a function of flow time. Centre: neck radius. Right: curvature profile at stopping. The result complements the frozen-profile hidden-term calculation: A′ is not a full-PDE coercive mechanism.

Proposition XIII.1 (Hidden-term and Ricci–DeTurck obstruction). The forcing $h = 2\lambda_1|\mathrm{Rm}|g$ gives a negative cubic contribution in the homogeneous Berger reduction. In the warped-S^3 sector its scalar-curvature evolution contains the additional term $-4\lambda_1\Delta|\mathrm{Rm}|$, and frozen neck profiles show that this derivative term is the same order as the retained algebraic term. A Ricci–DeTurck warped-flow diagnostic further shows that the engineered forcing delays and relocates curvature concentration rather than removing it. Therefore the Berger coercivity of A′ does not constitute evidence for the full pointwise coercive inequality.

XIV. FULL PDE OBSTRUCTION

The positive Berger theorem and the engineered cubic A′ both use homogeneity. The warped-S^3 diagnostics show concretely what homogeneity removes. In the full curvature equation for $\partial\tau g = -2\,\mathrm{Ric} + h$, $C^h(\mathrm{Rm})$ includes second derivatives of h and lower-order terms of type Rm * h. For $h = 2\lambda_1|\mathrm{Rm}|g$, the derivative contribution $-4\lambda_1\Delta|\mathrm{Rm}|$ is the same order as the algebraic cubic on a neck. The Ricci–DeTurck run adds the dynamic fact: the engineered term delays and relocates concentration, but does not remove it. This is exactly where finite-dimensional reductions can mislead.

A. Global volume normalization

The standard normalized Ricci flow in dimension three is $\partial\tau g = -2\,\mathrm{Ric} + (2/3)\bar{R}g$, where $\bar{R} = (1/\mathrm{Vol}(M,g))\int_M R\, d\mu_g$. This preserves total volume but is **not** pointwise coercive: a local curvature spike can satisfy $|\mathrm{Rm}|(x) \gg |\bar{R}|$. Then $\bar{R}g$ cannot dominate the local cubic reaction term.

B. Pointwise curvature-driven trace forcing

The pointwise analogue $\partial\tau g = -2\,\mathrm{Ric} + (2/3)R(x)g$ has the correct local homogeneity in positive-curvature homogeneous regimes. But now $h = (2/3)Rg$ depends on second derivatives of g, and the variation of curvature produces terms involving $\nabla^2 R$, Rm * R, ∇R * ∇g schematically.

The homogeneous Berger reduction hides these terms because all spatial derivatives vanish.

C. The correct open problem

> **Open Problem.** *Find a geometrically justified trace/expansion forcing, possibly combined with bounded shape filtering and Planck-boundary constraints, whose full curvature-level contribution satisfies a pointwise coercive inequality of the form (11), or prove that no such local tensorial forcing exists under stated regularity and covariance assumptions.*

XV. FORBIDDEN OVERCLAIMS

For clarity we record claims that should not appear in a defensible version of the programme.

(23) **Do not write:** the Hopf fibration removes singularities. **Write instead:** Hopf/Berger geometry separates fibre collapse from volume curvature blow-up.

(24) **Do not write:** dark energy is fibre curvature. **Write instead:** a dark-energy-like sector may be modelled as trace/volume compensation.

(25) **Do not write:** dark matter is solved. **Write instead:** dark-matter-like clustering can be tested later as quasi-local support; it is not used in the first theorem.

(26) **Do not write:** the Big Bang is Perelman's neck. **Write instead:** backward reconstruction may approach a boundary or neck-like limit of the coarse-grained description.

(27) **Do not write:** bounded spectral filtering prevents singularities. **Write instead:** bounded filtering is legitimate forcing but needs curvature-level coercivity.

(28) **Do not write:** the Planck boundary proves quantum gravity. **Write instead:** a self-adjoint Planck endpoint gives an effective no-flux boundary under stated assumptions.

(29) **Do not write:** normalized Ricci flow proves cosmological regularity. **Write instead:** it gives a positive homogeneous model and an open PDE problem.

XVI. ARTICLE-LEVEL CONCLUSIONS

The first-stage spectral-filter framework identified the true analytic bottleneck: bounded forcing is not enough; one needs a negative cubic curvature contribution or a mechanism that replaces the region where the cubic blow-up would occur. The Hopf/Berger test sharpens this statement. The clean result is mixed.

First: Fibre-only compensation is a no-go for the main singular mode. It can affect c, the Hopf fibre scale, but the unforced Ricci-flow singularity in the Berger family is the round/base/volume collapse a → 0. Fibre collapse c → 0 at fixed a has bounded curvature.

Second: The trace/expansion sector is essential. The curvature-driven trace flow $\partial\tau g = -2\,\mathrm{Ric} + (2/3)Rg$ preserves Berger volume, drives $u = (c/a)^2$ toward 1, and gives $|\mathrm{Rm}|^2 \le C(u_0)R_{\mathrm{eff}}^{-4}$. With a Planck lower boundary $R_{\mathrm{eff}} \ge \ell P$, this becomes $|\mathrm{Rm}|^2 \le C(u_0)\ell P^{-4}$.

Third: The engineered cubic forcing $h = 2\lambda_1|\mathrm{Rm}|g$ succeeds only in homogeneous reduction. In the warped-S^3 test the hidden derivative term $-4\lambda_1\Delta|\mathrm{Rm}|$ is the same order as the retained algebraic cubic term. Hence A' is not a full-PDE coercive mechanism.

Fourth: The positive theorem is finite-dimensional and boundary-based. The clean mechanism presently established is not a smooth coercive forcing but a trace/shape-controlled invariant region with a Planck no-flux lower boundary.

The version-2.0 programme is therefore:

Ricci shrinkage + trace/ADM expansion + Hopf shape/fibre sector + Planck boundary.

The Boris/Claude Berger diagnostics are therefore not a competing conclusion. They are the negative half of the result: they prove that fibre compensation alone is insufficient and motivate the trace/expansion term used in the positive theorem.

This is stronger than the fibre-only proposal and more honest than a singularity-resolution claim. It yields one clean homogeneous theorem, several no-go statements, a nonhomogeneous hidden-term obstruction, a Ricci–DeTurck dynamic failure test for the engineered cubic, and a precise full-PDE open problem.

APPENDIX A: DERIVATION DETAILS FOR THE BERGER FORMULAS

With $d\sigma_i = -2\varepsilon_{ij}{}^k\,\sigma_j \wedge \sigma^k$ and $\theta^1 = a\sigma^1$, $\theta^2 = a\sigma^2$, $\theta^3 = c\sigma^3$, the structure equations become:

$$d\theta^1 = -2/c \cdot \theta^2 \wedge \theta^3, \quad (A1)$$

$$d\theta^2 = -2/c \cdot \theta^3 \wedge \theta^1, \quad (A2)$$

$$d\theta^3 = -2c/a^2 \cdot \theta^1 \wedge \theta^2. \quad (A3)$$

Solving Cartan's first structural equations and then computing curvature gives sectional curvatures:

$$K_{12} = (4 - 3u)\,/\,x, \quad K_{13} = K_{23} = u\,/\,x. \quad \textit{(A4–A5)}$$

Ricci eigenvalues: $r_1 = K_{12} + K_{13} = 2(2-u)/x$, $r_3 = K_{13} + K_{23} = 2u/x$, reproducing (31). Also $|Rm|^2 = 4(K^2{}_{12} + K^2{}_{13} + K^2{}_{23}) = 4(16-24u+11u^2)/x^2$.

APPENDIX B: FORCING CONTRIBUTIONS BY EULER HOMOGENEITY

The curvature scalar $K_B(x,z) = 4(16x^2 - 24xz + 11z^2)/x^4$ is homogeneous of degree −2 in (x,z).

For any isotropic trace forcing $h = 2\lambda g$: $\dot{x} = 2\lambda x$, $\dot{z} = 2\lambda z$. Euler's identity gives $\dot{K}_B = 2\lambda(x\partial_x + z\partial_z)K_B = -4\lambda K_B$.

For fibre forcing $\dot{z} = \mu$, $\dot{x} = 0$: $\dot{K}_B = \mu\,\partial_z K_B = \mu\cdot 8(11z - 12x)/x^4$.

APPENDIX C: NUMERICAL VALIDATION PROTOCOL

The theorems above are analytic. Numerical experiments should therefore be used only as checks. A minimal validation suite should integrate:

$$\dot{x} = -8 + 4u + H_x(x,u), \quad \dot{z} = -4u^2 + H_z(x,u) \quad \textit{(C1–C2)}$$

for the following cases:

(30) unforced flow $H_x = H_\beta = 0$;
(31) constant trace forcing $H_x = 2\Lambda x$, $H_\beta = 2\Lambda z$;
(32) fibre forcing $H_x = 0$, $H_\beta = \mu$;
(33) normalized trace forcing $H_x = (2/3)Rx$, $H_\beta = (2/3)Rz$;
(34) Planck-reflected flow with $R_{eff} \geq \ell_P$.

The monitored quantities should be u, V, K_B, and $K_B\ell^4{}_P$. A numerical statement that contradicts volume preservation or (62) is a diagnostic error, not a new effect.

The representative validation suite used for the figures gives the following qualitative outcomes.

Run	Forcing	Observed outcome
E1	none	round-point blow-up; $u \to 1$; $\tau^* \approx 0.137$ for $(a,c)=(1,0.2)$
E2	$h = 2\Lambda g$, $\Lambda = 1$	still blows up; quadratic damping is insufficient
E2b	$h = 2\Lambda g$, $\Lambda = 5$	runaway expansion, $\lvert Rm\rvert^2 \to 0$ by escape, not coercive control
E3	fibre $h_{33} = 30(0.25 - z)+$	prevents c-collapse but u grows; blow-up remains
E4/E5	reflecting Planck barrier $a \geq 0.5$	bounded; sup $\lvert Rm\rvert^2 \ell_P^4 \approx 56$ in the test run
D1	coercive diagnostic	round ratio gives $4/\sqrt{3}$; fibre sign flip at $u = 12/11$

Table C1. Numerical checks are used only as diagnostics. The analytic statements are the curvature formulas, no-go computations, and the trace-compensation bound.

Appendix D: Frozen-Profile and Ricci–DeTurck Diagnostics

This appendix records the finite calculation behind Section XIII. For the warped metric $g = ds^2 + \psi^2 gS^2$, the round sphere $\psi = \sin s$ gives $K_0 = K_1 = 1$, $R = 6$, and $\lvert Rm\rvert^2 = 12$. This fixes the normalization.

The conformal scalar-curvature variation in dimension three is $\delta R = -2\varphi R - 4\Delta\varphi$. With $\varphi = \lambda_1 \lvert Rm\rvert$, the full contribution of $h = 2\lambda_1 \lvert Rm\rvert g$ contains both $-2\lambda_1 \lvert Rm\rvert R$ and $-4\lambda_1 \Delta \lvert Rm\rvert$. The Berger model sets the second term to zero because $\lvert Rm\rvert$ is spatially constant.
The frozen-neck profile test compares the magnitude of the omitted derivative term with the retained algebraic term. The representative ratios 3.18, 1.49, 0.97, 0.80, 0.77 show that the omitted term remains order one relative to the kept term. This ratio calculation is a scale diagnostic, not a complete dynamic solver.
The Ricci–DeTurck diagnostic uses a rotationally symmetric parabolic gauge with round background $g_bar = dx^2 + \sin^2 x\, gS^2$. It validates the round solution and the unforced neckpinch, then tests A'. The observed behaviour is that A' holds the neck open and delays concentration, but the curvature peak is displaced rather than removed. Thus the dynamic test supports the analytic hidden-term obstruction.

APPENDIX E: OPEN PROBLEMS

(35) **OP1.** Prove short-time existence for Ricci–DeTurck flows with combined trace, spectral, boundary, and nonlocal forcing.

(36) **OP2.** Determine whether a geometrically natural pointwise trace forcing can satisfy the coercive inequality without uncontrollable $\nabla^2 R$ terms.

(37) **OP3.** Formulate an anisotropic Planck-boundary Wheeler–DeWitt problem for Berger variables (V, u), including the correct minisuperspace measure and self-adjoint boundary form.

(38) **OP4.** Extend the Berger calculation to non-left-invariant perturbations on S^3 and determine whether trace compensation stabilizes only the homogeneous mode or also low tensor harmonics.

OP4′. Turn the Ricci–DeTurck warped-S^3 diagnostic into a fully convergence-tested neckpinch solver, including pole regularization and comparisons with Lagrangian or arclength-gauge schemes.

(39) **OP5.** Derive an effective stress tensor from Hopf connection curvature using Riemannian submersion or Kaluza–Klein methods and test its curvature-level sign.

(40) **OP6.** Combine bounded tensorial spectral filtering of shape modes with trace compensation and prove an invariant region in a finite tensor-mode truncation.

(41) **OP7.** Decide whether a dark-matter-like clustering sector helps curvature control or inevitably worsens local concentration in the Ricci-type comparison estimate.

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